

Sample Project — Mixed-Use Block

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Architectural Design Brief

Sample document — testing only. This file exists to give Castor users realistic prose to query in Ask and to propose changes against in Modify. Codes, dimensions, and material values are written in EU (Eurocode / Spanish CTE) format and **may be technically wrong** — spotting and flagging discrepancies is part of the demo, not a bug. Do not use any value here as a basis for real design.

1. Scope

This brief covers the architectural intent for a four-storey mixed-use building comprising ground-floor retail, three storeys of residential apartments above, and a partial basement for plant and storage. It is issued at Design Development for coordination with the structural, MEP, and FF&E packages. Drawings, schedules, and specifications cited here are referenced by document number; nothing in this brief shall be taken as superseding a stamped drawing.

2. General requirements

- All internal heights shall be **2.70 m clear** in residential units and **3.20 m clear** in retail units, measured from FFL to underside of slab.
- Accessible routes shall comply with **CTE DB-SUA 9** with a clear width of not less than **1.20 m** through corridors and **0.85 m** through doors.
- Acoustic separation between dwellings shall achieve a minimum airborne **DnT,A ≥ 50 dB** and impact **L'nT,w ≤ 60 dB**.
- Fire compartmentation shall provide **EI 60** between dwellings and **EI 90** between dwellings and the basement.

3. Partition build-ups

Tag	Build-up	Total thickness	Acoustic (Rw)	Fire
P1	2 × 15 mm gypsum / 70 mm steel stud / 60 mm mineral wool / 2 × 15 mm gypsum	130 mm	52 dB	EI 60
P2	1 × 15 mm gypsum / 70 mm steel stud / 50 mm mineral wool / 1 × 15 mm gypsum	115 mm	45 dB	EI 30
P3	200 mm CMU plastered both faces	230 mm	55 dB	EI 120

P1 separates dwellings. P2 is internal to dwellings. P3 separates the basement plant rooms from circulation.

4. Finishes by room type

Room	Floor	Walls	Ceiling
Living / Dining	Engineered oak, 14 mm	Painted gypsum, matt	Painted gypsum, matt

Bedroom	Engineered oak, 14 mm	Painted gypsum, matt	Painted gypsum, matt
Kitchen	Porcelain tile, 600 × 600 mm	Painted gypsum + splashback tile to 1.5 m	Painted gypsum
Bathroom	Porcelain tile, 300 × 600 mm	Tiled to underside of slab	Moisture-resistant gypsum, painted
Corridors (common)	Polished concrete with sealer	Painted gypsum	Suspended acoustic tile
Retail	Polished concrete (tenant fit-out base)	Painted gypsum	Exposed services with feature paint
Plant / storage	Sealed concrete	Painted CMU	Exposed slab

5. Doors and windows

- **Apartment entrance doors:** solid core, 45 mm, fire-rated EI 60-Sm with self-closer and intumescent seals. Lock to BS 8621.
- **Internal apartment doors:** hollow-core, 40 mm, painted MDF facings.
- **Bathroom doors:** hollow-core with privacy lock, undercut 10 mm for ventilation.
- **External windows (residential):** triple-glazed argon-filled, $U \leq 1.1 \text{ W/m}^2\text{K}$, openable casement on at least one leaf per room.
- **Retail shopfronts:** structural glazing, single full-height span, manual recessed entrance door.

6. FF&E (Design Development scope)

The architectural package coordinates the loose furniture and built-in joinery shown on the room datasheets (drawings ARCH-FF&E-101 to 305). Loose furniture is selected by the interior team but subject to the architectural acoustic and fire constraints in §2.

Per typical dwelling unit:

- Living / Dining: 1× three-seat sofa, 1× armchair, 1× side table, 1× dining table for four, 4× dining chairs, 1× sideboard.
- Master bedroom: 1× double bed (1.50 × 2.00 m), 2× bedside tables, 1× wardrobe (built-in, 2.40 m wide, full-height).
- Secondary bedroom: 1× single bed (0.90 × 2.00 m), 1× desk, 1× chair, 1× wardrobe (built-in, 1.80 m wide).
- Kitchen: built-in U-shape with 600 mm dishwasher, 600 mm oven, induction hob, integrated extractor.
- Bathroom: ceramic WC, 1.20 × 0.80 m walk-in shower with glass screen, 800 mm vanity unit.

All built-in joinery shall be flush-fronted MDF, painted to match adjacent wall finish unless noted otherwise on the room datasheets.

7. Open coordination items

The following items are flagged for resolution in the next coordination cycle and may not be reflected in the current IFC export:

- Ceiling plenum depth in retail units pending MEP riser layout — value shown on drawings is provisional.
- Acoustic treatment of the lift lobby is under review; the current brief assumes the corridor specification applies.
- FF&E for retail tenant zones is excluded from this package and delivered under the tenant fit-out scope.

End of document. Drawings, schedules, and product datasheets referenced above are issued separately.

